Static and kinetic friction

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Distinguish adjustable static friction from sliding friction.

SUCCESS CRITERIA - I can...

1. Compare a push with the static-friction limit.
2. Find actual friction for rest or sliding.
3. Explain friction direction using relative slipping.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A runner pushes the ground backward with 50 N. Name the partner force, its size and the object it acts on. Explain its role in the runner's motion.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $0 \le f_s \le \mu_s N$; $f_k = \mu_k N$.
s = static; k = kinetic; N = normal force. Static friction adjusts up to its maximum. Kinetic friction applies during sliding.

Worked reasoning: A box remains still while pushed with 3 N. Static friction is 3 N opposite the push, provided its maximum is at least 3 N. Static friction need not equal its maximum.

Your turn: Draw the horizontal forces. Then redraw for a 5 N push that is still below the maximum.



Box remains at rest. Redraw for a 5 N push.

Marzano check - circle: ① ② ③

03 Below the limit

The box stays at rest with an 8 N push right. Maximum static friction is 15 N. Find actual friction and net force.



The dot represents the object. At rest; static limit 15 N

Marzano check - circle: ① ② ③

04 Crossing the limit

The push on the box in question 3 rises to 18 N. Once it slides right, kinetic friction is 12 N. Complete the diagram and find the net force. Explain why you use 12 N rather than the 15 N static limit.



Marzano check - circle: ① ② ③

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Continue down each column.

05 Can friction help?

A runner's foot pushes backward against the ground without slipping. Draw the friction force the ground exerts on the foot. Explain why "friction always points backward" is not a reliable rule.

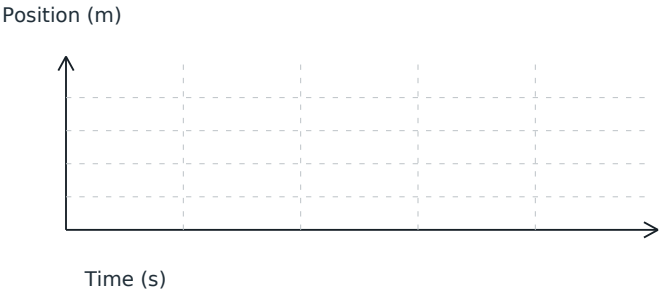


Foot Ground

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A cart starts at 4 m and moves right at 6 m/s for 3 s. Predict final position. Allowed endpoints are 20 to 24 m; state this interval and one assumption.
Label the model with the given positions and any prediction.



Marzano check - circle: 1 2 3

07 Exit check

A box has a maximum static friction of 20 N. In one trial, a 12 N push leaves it at rest. In a new trial, a 25 N push starts it sliding with 16 N kinetic friction. Find actual friction in the first trial and net force in the second.

Trial	Push (N)	Friction (N)	Net force (N)
At rest	12	____	____
Sliding	25	16	____

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A box has maximum static friction 8 N and kinetic friction 6 N. Predict the behavior for pushes of 4, 8 and 10 N. Draw the sliding case.



Add and label force arrows.

Draw the sliding case, with actual friction.

Marzano check - circle: 1 2 3